

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2008

BIOLOGY
HIGHER 2

9747/02

Paper 2 Core Paper

15 AUGUST 2008

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on this answer booklet.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

Write your answers on the lined paper provided in the question paper.

At the end of the examination, circle the number of the Section B question you have answered in the grid opposite.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
8 / 9	
Total	100

This question paper consists of **20** printed pages including **4** lined pages.

[Turn over

- 1 Fig. 1.1, 1.2 and 1.3 are electron micrographs taken from different cells.

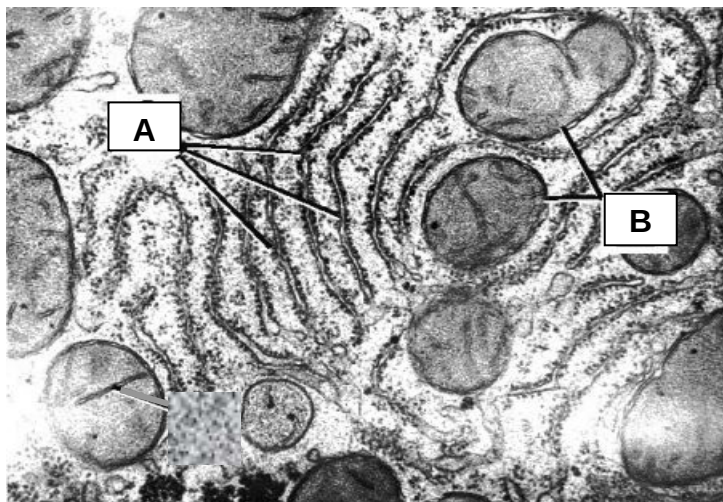


Fig 1.1

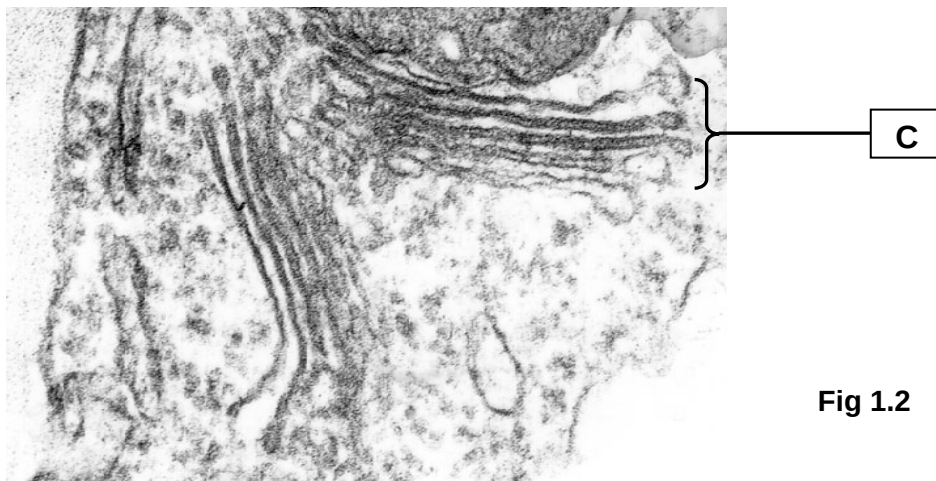


Fig 1.2

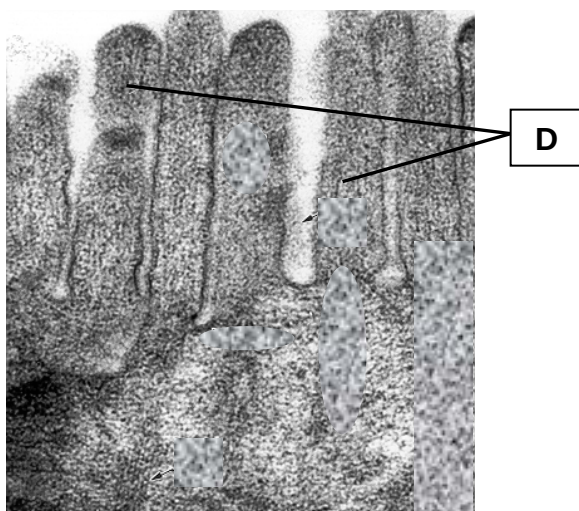


Fig 1.3

(a) Name the structures labeled A to D.

A _____

C _____

B _____

D _____ [2]

(b) Describe briefly how the two structures, **A** and **C**, are related in the cell.

[3]

(c) With reference to Fig. 1.1, explain the significance of the numerous **B** found adjacent to **A**.

[1]

(d) (i) State the function of structure **D**.

[1]

(ii) Suggest where such structures may be found in the human body.

[1]

Fig. 1.4 shows a stalked particle that is found in internal membranes in cells. These structures are responsible for the production of ATP that is driven by the proton (H^+) gradient that occurs across such membranes.

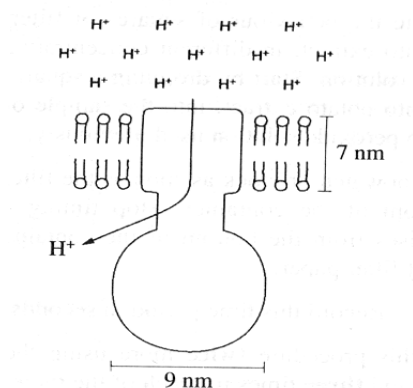


Fig 1.4

(e) State the precise locations of such particles in a general plant cell.

[2]

(f) Describe how the structure of the stalked particle is related to its function.

[3]

[Total: 13]

- 2 In an experiment to study the effect of heat treatment on the digestibility of protein substrate and the effect of raw bean extract on protease activity, various reaction mixtures were prepared and were incubated for 30 minutes.

The protein concentration of each reaction mixture at the beginning and at the end of the incubation period was determined by the colorimetric method which measures colour intensity of these reaction mixtures. The results were shown in table 2.1.

Table 2.1

Incubation period /min	Colour intensity of the reaction mixture / arbitrary unit		
	Tube A	Tube B	Tube C
	Protease + heated protein substrate	Protease + unheated protein substrate	Protease + unheated protein substrate + raw bean extract
0	10	10	10
30	4	7	9

Fig. 2.1 shows a standard graph obtained by using colorimetric method for determining concentration of protein solutions.

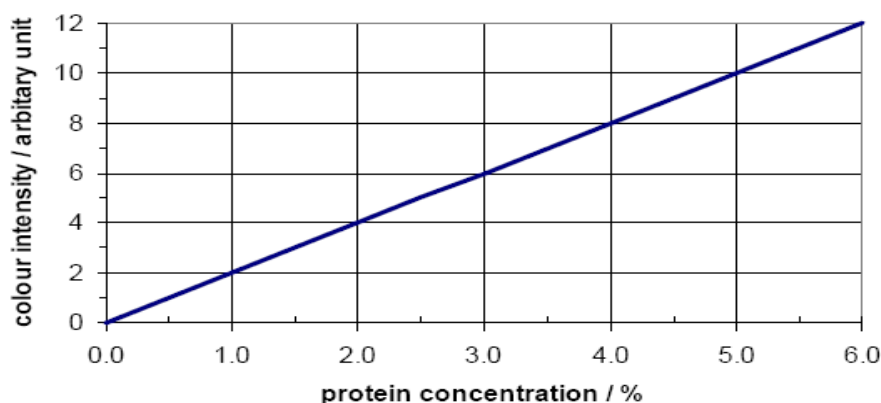


Fig. 2.1

(a) With reference to Table 2.1 and Fig 2.1,

- (i) Determine the protein concentration of the three tubes at time 0 min.

_____ [1]

- (ii) Explain which protein substrate is more digestible by protease.

_____ [2]

- (b) Based on your knowledge of protein structure, explain the effect of heat treatment in the digestibility of this protein substrate.

[3]

- (c) Suggest the effect of raw bean extract on protease activity.

[1]

- (d) Fig 2.2 shows the structure of a biochemical molecule, **W**.

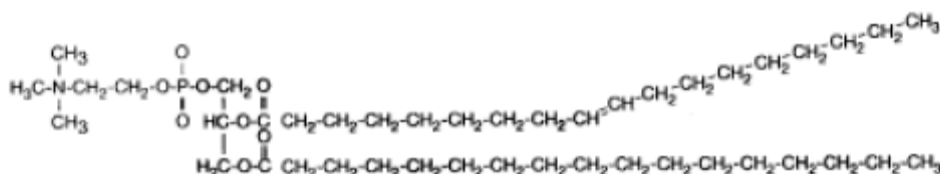


Fig 2.2

- (i) Identify the molecule **W**.

[1]

- (ii) Proteins can associate with this molecule in the cell membrane. Based on your knowledge on the structure of amino acids/proteins, give a reason for this observation.

[2]

- (iii) Membrane lipids in reindeer hooves need to remain fluid at low temperatures. Based on the structure of molecule **W**, predict the composition of such membrane lipids.

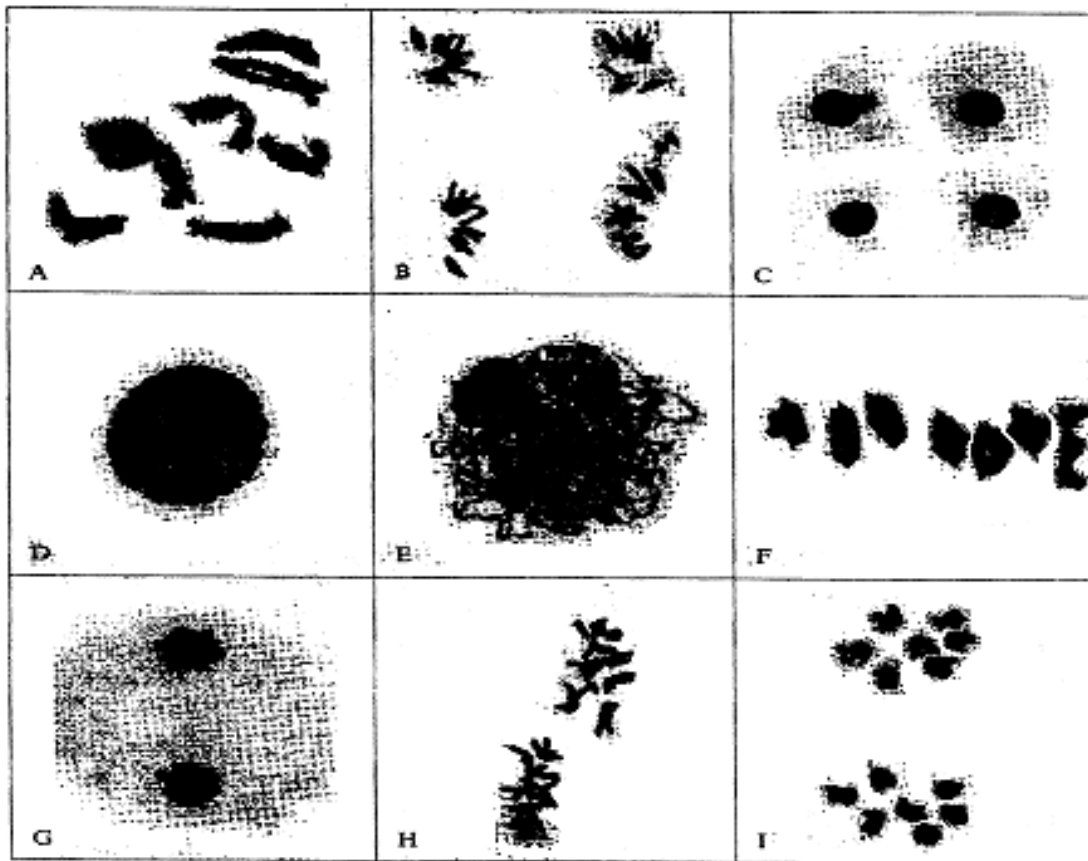
[1]

- (iv) The feather of certain aquatic birds such as ducks are covered with a protective layer of wax. Based on this information, predict why oil spills are so harmful to these birds.

[1]

[Total: 12]

- 3 The micrographs below show nuclei of cells at various stages during nuclear division in a flowering plant (Fig. 3.1).



Adapted from: *Introducing Genetics*, Jones and Karp, John Murray, 1988.

Fig. 3.1

- (a) Micrograph D shows the first stage in the sequence. In the boxes below, arrange the other letters shown on the micrographs to indicate the correct chronological sequence for this type of nuclear division.

D								
---	--	--	--	--	--	--	--	--

[1]

- (b) State the diploid number of this plant. With reference to **two** stages (name the stages) from the photomicrographs, explain how you arrived at your answer.

[3]

- (c) During the stage shown in **H**, independent assortment will result in an increased variety of gametes if a particular process has taken place at an earlier stage. Name this process and state when this occurs during cell division.

Process: _____

Stage: _____ [2]

- (d) With reference to micrograph **F** in Fig. 3.1, fill in the numbers of the following:

No. of sets of chromosomes _____ No. of centromeres _____

No. of chromatids _____ No. of telomeres _____

No. of DNA strands _____ [2]

- (e) Distinguish between the terms centriole and centromere.

_____ [3]

- (f) If centromeric DNA is deleted from a chromosome, suggest how this will affect the process of mitosis.

_____ [3]

[Total 14]

- 4 (a) Explain how the promoter of a eukaryotic gene influences transcription.

[3]

- (b) Fig. 4.1 shows the binding of a general transcription factor to part of the promoter. This will help to recruit RNA polymerase to the promoter.

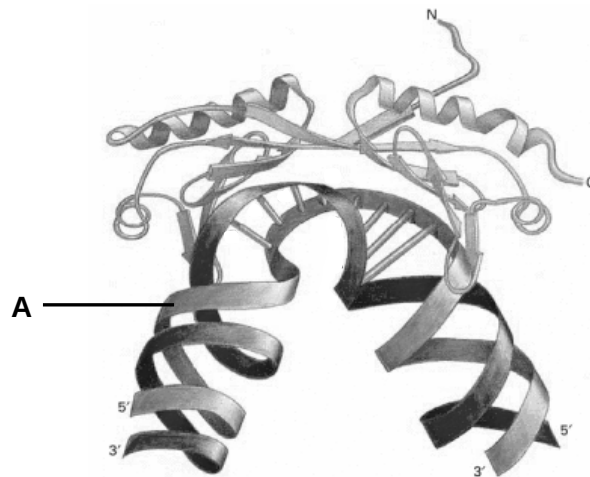


Fig. 4.1

- (i) With reference to Fig. 4.1, give **two** reasons why the structure labelled **A** is part of the promoter and not the transcription factor.

[2]

- (ii) Suggest why a substitution mutation at the promoter may result in the silencing of the gene controlled by the promoter.

[2]

(d) Describe how distal control elements can help increase the rate of transcription of a gene.

[3]

(e) Even though the control of gene expression occurs mostly at the level of transcription, control of gene expression also occurs at the level of post-translational modification.

(i) Name an example of post-translational modification of a **specific** protein.

[1]

(ii) Suggest the advantage of control of gene expression at the level of translation over control of gene expression at the level of transcription.

[1]

[Total: 12]

- 5 In *Drosophila* flies, a gene for wing shape found on chromosome 2 has two alleles – allele **A** for curly wing, which is dominant to allele **a** for normal wing.

Female flies are homogametic while males are heterogametic. The gene for *Drosophila* eye colour is located on the X chromosome and has two alleles – the allele for red eyes being dominant to the allele for white eyes.

- (a) (i) Write the genotypes of the two flies in the table below.

	Fly 1	Fly 2
Sex of fly	Female	Male
Eye colour	White	Red
Wing shape	Curly wing, heterozygous	Curly wing, heterozygous
Genotype		

[1]

- (ii) Draw a genetic diagram showing the phenotypes of the offspring and the expected ratio of phenotypes when these two flies are crossed.

[4]

In a separate experiment, when a white-eyed and yellow-bodied (homozygous recessive) female was crossed with a red-eyed and gray-bodied (dominant) male, all the F1 females have gray body and red eyes whereas all the F1 males expressed white eyes and yellow body. In the F2 generation, 98.7% of the total offspring showed the parental phenotypes - yellow-bodied, white-eyed flies and gray-bodied, red-eyed flies. The remaining 1.3% of the flies were either yellow bodies with red eyes or gray bodied with white eyes.

(b) (i) State the possible explanation for the results obtained for F1.

_____ [1]

(ii) Given the percentages of the different phenotypes obtained in the F2 generation, state the information regarding the two genes investigated that can be calculated.

_____ [1]

(iii) State the process and stage during the formation of gametes in the flies that results in the appearance of the above said 1.3% of the offspring.

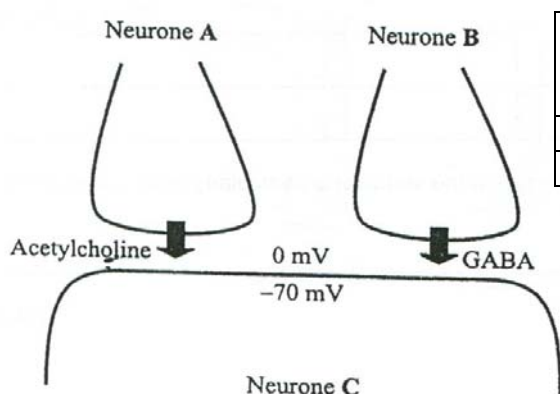
_____ [1]

(c) From the results obtained in the above cross, would you conclude that the inheritance of eye and body colour shows continuous or discontinuous variation? Explain your answer.

_____ [2]

[Total: 10]

- 6 Fig. 6.1 shows dendrites from each of two neurones, **A** and **B**, forming synapses with a third neurone, **C**. Concentrations of some ions are also shown.



	Ion concentration / nmol dm ⁻³		
	Na ⁺	K ⁺	Cl ⁻
Outside neurone C	145	12	120
Inside neurone C	12	155	5

Fig 6.1

- (a) (i) Acetylcholine increases the permeability of the postsynaptic membrane to Na⁺ ions. Explain how acetylcholine released from neurone **A** generates an action potential in neurone **C**.

[4]

- (ii) Neurone **B** releases a substance called GABA as its neurotransmitter. GABA increases the permeability of the postsynaptic membrane to Cl⁻ ions. Suggest how impulses in neurone **B** would reduce the chances of an action potential in neurone **C**.

[2]

- (iii) Suggest a benefit of neurone **C** synapsing with more than one neurone.

[1]

- (b) (i)** The acetylcholine receptor is a type of G-protein-linked receptor. When acetylcholine binds to it, a membrane-bound enzyme is activated. Name the enzyme.

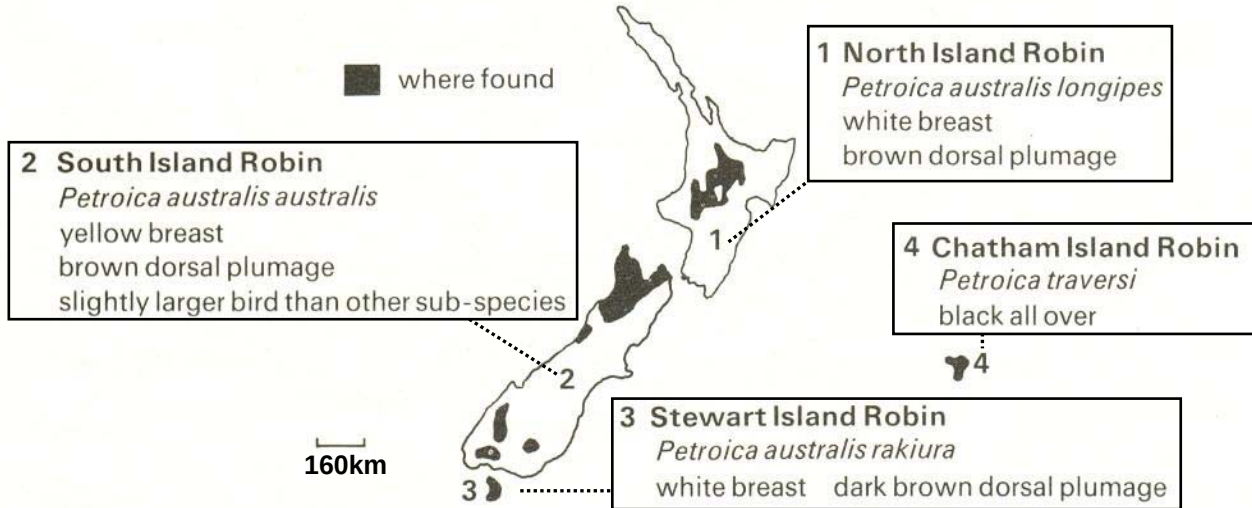
_____ [1]

- (ii)** Explain the role of the enzyme named in (b)(i) in signal transduction that leads to a response in the postsynaptic neurone.

_____ [2]

[Total: 10]

- 7 In New Zealand today, there are two species and three sub-species of native bush robins. It is believed that the robins evolved from a common ancestral stock, members of which flew from Australia across the Tasman Sea and became established in New Zealand over a million years ago. This ancestral form is considered to be similar to the present day Australian flame robin, *Petroica multicolor*, a bird with a brightly coloured red breast. The New Zealand birds do not have this red colour. Some characteristics of the birds and their distributions are shown below:



- (a) State the scientific name of the two species of native bush robins in New Zealand.

_____ [1]

- (b) Explain why the robins in locations 1, 2 and 3 are similar but different from those in location 4.

_____ [4]

- (c) Describe how the molecular clock can be used to date evolutionary events, such as when the two species of native bush robins in New Zealand diverged.

_____ [2]

- (d) The red breast trait is present in Australian robins. Account for the absence of this trait in New Zealand robins.

[2]

[Total: 9]

Section B

Answer **one** question.

Write your answers in the lined pages provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a) Compare and contrast the viral genome and the eukaryotic genome. [8]
- (b) Explain the role of the F plasmids in bacterial conjugation. [8]
- (c) Suggest the advantages of conjugation as compared to transduction. [4]
- 9 (a) Explain the principles of homeostasis in terms of receptors, effectors and negative feedback. [7]
- (b) Compare and contrast competitive and non-competitive inhibitors and their effects on the rate of enzyme activity. [7]
- (c) Suggest similarities between the interaction of substrate with an enzyme and the interaction of ligand with a receptor. [6]

